

ME 300:
Engineering Thermodynamics

Example 2:
Ideal Gas Equation
of State: P - v - T

Example: Ideal gas

- I have two containers with the same volume and at the same temperature inside. One has 10 grams of Helium in one and 10 grams of Argon in the other. Which one has a higher pressure?

- Molecular weights:

- Ar = 40 kg/kmol
- He = 4 kg/kmol

m

$$T_{He} = T_{Ar}$$

$$V_{He} = V_{Ar}$$

$$M_{He} = M_{Ar}$$



Which vessel has a higher pressure?

- a) Helium
- b) Argon

Method #1

$$PV = NR_uT \quad \rightarrow N \sim [\text{mol}]$$
$$R_u = 8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}}$$

$$V_{\text{He}} = V_{\text{Ar}}$$

$$T_{\text{He}} = T_{\text{Ar}}$$

$$P = \frac{NR_uT}{V} \Rightarrow N?$$

$$P_{\text{He}} = 10 P_{\text{Ar}}$$

$$\text{Argon: } N_{\text{Ar}} = \frac{M_{\text{Ar}}}{M_{\text{Ar}}} = \frac{10 \text{ kg}}{40 \text{ kg/kmol}} = 0.25 \text{ kmol}$$

$$\text{Helium: } N_{\text{He}} = \frac{M_{\text{He}}}{M_{\text{He}}} = \frac{10 \text{ kg}}{4 \text{ kg/kmol}} = 2.5 \text{ kmol}$$

Method #2

$$P\check{V} = \check{M}RT\check{V}$$

$$\hookrightarrow R_{Ar} = \frac{R_u}{\eta_{M_{Ar}}} = 208 \frac{\text{J}}{\text{kg}\cdot\text{K}}$$

$$R_{He} = \frac{R_u}{\eta_{M_{He}}} = 2080 \frac{\text{J}}{\text{kg}\cdot\text{K}}$$

$$\left. \begin{array}{l} P = \frac{\check{M}RT\check{V}}{\check{V}} \\ P_{He} > P_{Ar} \\ R_{He} > R_{Ar} \end{array} \right\}$$